CLASS PARTICIPATION COVERSHEET

Unit of Study: **MIT653 Research Methods**

Assignment Title: **Class Participation – Week 02**

Trimester and Year: **Trimester 2, 2026**

Lecturer Name: **Dr. Sudath Heiyanthuduwage**

DECLARATION

The work contained in this assignment is my own work and has not been copied from other sources or been previously submitted, unless otherwise specifically acknowledged.

I understand that failure to comply can lead to severe penalties.

Student Identification: ____________________________________

Student Name: ____

Date Submitted: ____dd____mm____yy

Signed: _____________________ Date: ____dd____mm____yy

EXAMINER ONLY

Due Date: ____dd____mm____yy

MARK: __________________________

FINAL MARK: ____________________

Section A:

1. Describe the steps involved in drafting a research question.

There are basically two main steps. First, you need to identify the research problem, this means picking a broad topic or area you want to look into, based on things like your academic background, personal interest, or trends happening in your field. Second, you take that broad problem and refine it into a clear and focused research question. This refining step should be guided by a literature review (to see what has already been studied) and by ethical considerations. In the end, the research question should be focused, specific, relevant, feasible, and significant.

2. Why is it important to have a clear and focused research question?

A clear and focused research question works like a guide for the whole research project. It tells you exactly what you are studying, which keeps the study from becoming too broad or confusing. It also helps set the scope of the research and gives a roadmap so the researcher knows what steps to take next, like which literature to review, what hypothesis to form, and what method to use for data collection. Without a focused question, it would be hard to know what data to even collect or how to answer it properly.

3. What are the key components of a good hypothesis?

A good hypothesis needs three main things. It should be specific, meaning it makes a clear prediction about the relationship between variables or the expected result. It should be testable, meaning it can actually be checked through real observation or experiments, and the data collected can show if it is supported or not. And it should be relevant, meaning it directly connects back to the research question and matches the goals of the study.

4. Discuss the role of literature in the process of drafting a research question and hypothesis.

Literature review plays a big role in both steps. When forming the research question, reviewing existing literature helps the researcher understand what is already known, so they can find a gap that has not been studied yet and turn a broad problem into a specific, focused question. When it comes to the hypothesis, literature gives the researcher a base of existing findings and theories, so the hypothesis is not just a random guess but a prediction that is supported by what past research has already shown. Basically, without reviewing the literature first, both the research question and hypothesis would lack direction and credibility.

5. What are the potential consequences of not defining a clear research problem, research question, or hypothesis before beginning a study?

If a study starts without clearly defining these things first, several problems can happen:

- The research can become unfocused, and the researcher might not know exactly what they are trying to find out.
- Time, money, and effort could be wasted, since planning helps allocate resources properly and without it, resources might be used the wrong way.
- The research design and data collection method might not match what the study is actually trying to test.
- The results might end up being irrelevant, not answerable, or not contribute anything meaningful to the field.
- It becomes harder to test or measure anything properly, since there is no clear, testable prediction to check against.

Section B: Case Study Evaluation

Case Study: Research Planning and Drafting Research Questions/Hypotheses

<u>Research Proposal 1</u>

Research Question: What is the relationship between mindfulness meditation and stress levels in working adults?
Hypothesis: Practicing mindfulness meditation will reduce stress levels in working adults.

Evaluation: Good

This is a good example. The research question is focused and specific, it clearly names the two variables (mindfulness meditation and stress levels) and the population (working adults). The hypothesis is also specific and testable, it makes a clear prediction (meditation reduces stress) that can actually be measured, for example by comparing stress levels before and after meditation. The hypothesis directly matches and answers the research question, so both work well together.

<u>Research Proposal 2</u>

Research Question: What are the factors that contribute to job satisfaction among employees in the healthcare industry?
Hypothesis: Employees who receive higher salaries are more likely to be satisfied with their jobs in the healthcare industry.

Evaluation: Bad (needs improvement)

The research question itself is fine, it's relevant and feasible. But there's a mismatch between the question and the hypothesis. The research question asks about "factors" (plural), suggesting the study wants to look at multiple things that affect job satisfaction, like workload, work environment, relationships with colleagues, etc. But the hypothesis only focuses on one single factor, salary. This makes the hypothesis too narrow compared to the scope of the research question, so it doesn't fully test what the question is asking.

Suggestion for improvement: Either narrow the research question to focus specifically on salary, for example: "Does salary level affect job satisfaction among healthcare employees?", or keep the broader research question and create multiple hypotheses to test different factors, such as salary, workload, and work environment, so that the hypotheses actually match the scope of the question.

<u>Research Proposal 3</u>

Research Question: How does parental involvement affect children's academic performance in middle school?
Hypothesis: Homeschooling leads to better academic performance in middle school children.

Evaluation: Bad

This is a bad example because the hypothesis does not match the research question at all. The research question is about parental involvement and how it affects academic performance, but the hypothesis is about homeschooling, which is a completely different concept. Homeschooling is a type of schooling method, while parental involvement is about how much parents are engaged, whether their child is homeschooled or not. So the hypothesis does not test what the research question is actually asking.

Suggestion for improvement: The hypothesis should directly relate to parental involvement, for example: "Higher levels of parental involvement are associated with better academic performance among middle school children." If the researcher specifically wants to study homeschooling, then the research question itself should be changed to something like: "How does homeschooling affect academic performance in middle school children compared to traditional schooling?"

Section C: Research Question, Hypothesis and Evaluation

Research Topic: *Impact of Artificial Intelligence on Cybersecurity*

Research Problem

Cybersecurity threats are becoming more complex and are evolving faster than traditional, rule-based security systems can handle. Organisations are increasingly integrating Artificial Intelligence (AI) and Machine Learning (ML) into their cybersecurity solutions to detect threats and respond to attacks more efficiently. However, the actual effectiveness of these AI/ML-based approaches, along with their limitations and practical challenges, still needs to be properly understood.

Research Question

"What is the impact of Artificial Intelligence and Machine Learning techniques on improving the efficiency and effectiveness of cybersecurity threat detection systems?"

Hypothesis

"The integration of Artificial Intelligence and Machine Learning techniques significantly improves the efficiency and accuracy of cybersecurity threat detection systems compared to traditional rule-based security approaches."

Evaluation

This research question and hypothesis match each other well. The research question directly asks about the impact of AI and ML on cybersecurity threat detection, and the hypothesis tests exactly that impact, in terms of efficiency and accuracy improvement. The scope of both is consistent, "AI and ML techniques" and "threat detection systems", so there is no mismatch of the kind seen in Proposals 2 and 3 in Section B. The hypothesis is also specific and testable, since efficiency and accuracy of AI-based systems can be measured and compared against traditional, rule-based approaches.

Github Repository:

[SS ADD - Screenshot of the Github repository containing this research question, hypothesis, and evaluation document]